

3.7 Interpreting Linear Models

Lesson Introduction

 Benchmark	MA.912.DP.2.4 Fit a linear function to bivariate numerical data that suggests a linear association, and interpret the slope and y -intercept of the model. Use the model to solve real-world problems in terms of the context of the data.		 Learning Targets	- I can interpret the correlation coefficient, slope, and y- intercept of a linear equation fit to real-world data. - I can use the linear equation to solve real-world problems in terms of the context of the data.
 Benchmark Coherence	Building On	Working Toward		
	MA.8.DP.1.2 Given a scatter plot within a real-world context, describe patterns of association.	MA.912.DP.2.8 Fit a quadratic function to bivariate numerical data that suggests a quadratic association, and interpret any intercepts or the vertex of the model. Use the model to solve real-world problems in terms of the context of the data. MA.912.DP.2.9 Fit an exponential function to bivariate numerical data that suggests an exponential association. Use the model to solve real-world problems in terms of the context of the data.		
 Essential Questions	- How can you interpret the correlation coefficient, slope, and y-intercept of a linear equation fit to real-world data? - How do you use the linear equation to solve real-world problems in terms of the context of the data?			
 Lesson Narrative	In this lesson, students will engage in a series of activities, each using bivariate data from a real-world context. For each set of data, students will write the equation of the line of best fit, interpret the meaning of key features (slope, y -intercept, correlation coefficient), determine the domain and range, use the model to make a prediction, and determine reasonableness of solutions in context.			
 Component Summary	3.7.1 Warm-Up (~5 min.)	Bell work on the line of best fit (<i>SE p. 144</i>)	3.7.3 Activity (10-12 min.)	Use bivariate data related to hurricanes in Florida to solve problems (<i>SE p. 147</i>)
	3.7.2 Activity (10-12 min.)	Use bivariate data related to high temperature and rainfall in Florida cities to solve problems (<i>SE p. 145</i>)	3.7.4 Activity (10-12 min.)	Use bivariate data related to the men's 100-meter dash to solve problems (<i>SE p. 148</i>)
	3.7.5 Wrap-Up (~5 min.)	Growth mindset reflection activity (<i>SE p. 149</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Glossary:</u> N/A		(Optional) Graphing utility	If utilizing as a station rotation, consider which stations may need more guidance, or groups that may need more guidance. This could translate to either pulling a small group to provide more guidance through the activities or setting up one of the stations as teacher-led.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may incorrectly interpret key features based on the context. - Students may make calculation errors when making predictions. - Students may try to make predictions from data points, rather than recognizing the linear model should be used to make predictions.	- This lesson is strictly activity-based with no formal instruction. Consider how you will group students together for each activity, based on their performance in previous lessons. - Consider using information from the previous two lessons to form a small group of students who are struggling working with bivariate numerical data. Facilitate these students as they work through the stations.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
 Differentiate Instruction	Group Presentations	MA.K12.MTR.7.1: Real-World Contexts
	Allow students to work in small groups for each of the lesson activities. At the conclusion of each activity, choose select groups to showcase their findings for specific questions, or parts of the activity to allow more groups to share out.	This lesson is comprised solely of problems that are presented in a real-world context. Students will connect mathematical concepts to everyday experience and use models and methods to understand, represent, and solve problems related to scatter plots and their data. Most of the contexts are also set in Florida to ensure connections to students' real-world experiences.
Lesson Notes:		

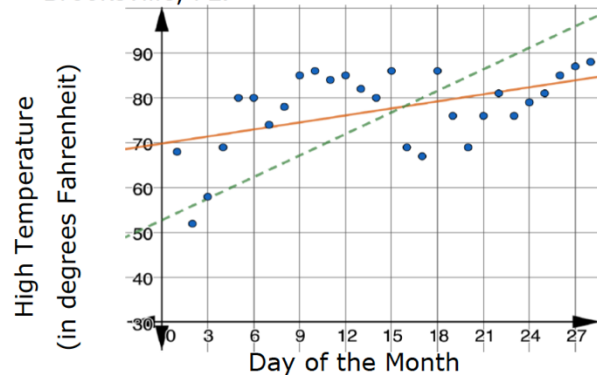
3.7.1 Warm-Up: Bell Work

Component Overview: Students will complete three questions about a scatter plot that represents a real-world context.



3.7.1 Warm-Up: Bell Work

- The scatter plot shows the relationship between the day of the month and the high temperature, in degrees Fahrenheit, for each day of February 2021 in Brooksville, FL.



- Which line fits the data better, the solid orange line or the dashed green line?

The solid line

- Explain your reasoning.

The solid line fits the data better because its slope and vertical intercept more closely resemble the trend of the data than the slope and vertical intercept of the dashed line.

- Use inequalities to represent an appropriate domain and range for the data represented.

Domain	$1 \leq x \leq 28$
Range	$51 \leq y \leq 89$



Watch student performance on this activity. If students struggle with these questions, they will likely struggle with the activities in this lesson. Use the data from this activity to determine which students will initially need support at the beginning of the activities.



Address any whole-group misconceptions demonstrated during this component with the class before releasing to stations.

3.7.2 Activity: Station 1: High Temperature and Rainfall

Component Overview: Students will analyze the data regarding high temperature and rainfall in 20 Florida cities to answer four, multipart questions.



3.7.2 Activity: Station 1: High Temperature and Rainfall

The table shows the mean high temperature in January, in degrees Fahrenheit ($^{\circ}\text{F}$), and the mean amount of rainfall per year, in inches (in.), for 20 cities in Florida.

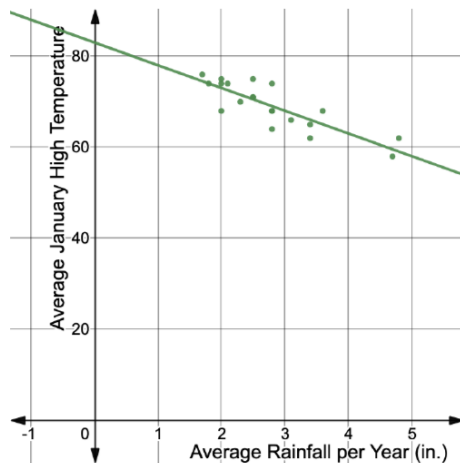
City	Mean Rain per Year (in.)	Mean High Temp in Jan. ($^{\circ}\text{F}$)	City	Mean Rain per Year (in.)	Mean High Temp in Jan. ($^{\circ}\text{F}$)
Alachua	3.6	68	Naples	1.7	76
Bradenton	2.5	71	Orlando	2.3	70
Clermont	2.8	68	Pensacola	4.7	58
Fort Lauderdale	2.5	75	Sarasota	2.5	71
Fort Myers	1.8	74	St. Augustine	3.1	66
Fort Pierce	2.1	74	Tallahassee	4.8	62
Gainesville	3.4	65	Tampa	2.0	68
Jupiter	2.8	64	West Palm Beach	2.8	74
Key West	2.0	74	Winter Garden	2.8	68
Miami	2.0	75	Yulee	3.4	62



Remind students that they may need to “bookmark” this page so they can refer to the data in the table, if needed, during the activity.

3.7.2 Activity: Station 1: High Temperature and Rainfall (continued)

The scatter plot shown represents the data from the table with a linear equation fit to model the relationship.



- The screenshot shows the slope, y-intercept, and correlation coefficient associated with the linear relationship modeled on the scatter plot.

$y_1 \sim mx_1 + b$	
STATISTICS	RESIDUALS
$r^2 = 0.7036$	e_1 plot
$r = -0.8388$	
PARAMETERS	
$m = -4.99154$	$b = 83.0265$

- Write the equation of the line of best fit shown.

$$y = -4.99154x + 83.0265$$



Students can use the screenshots in the workbook or go to the actual link from the digital area.



To assist students struggling to write the equation, consider asking the following questions:

- What do the values for m and b represent?
- How could you use that information to write the equation of the line?
- What form would be the easiest to write the equation based on this information?

3.7.2 Activity: Station 1: High Temperature and Rainfall (continued)

- b. Interpret the meaning of each value listed in the table.

Value	Meaning
slope, m	<i>The average January high temperature decreases by about 5 degrees for each 1 inch increase in average rainfall per year.</i>
y-intercept, b	<i>When the average rainfall is 0 inches, the average January high temperature is about 83 degrees.</i>
correlation coefficient, r	<i>Strong, negative linear association</i>



Listen for student responses to question 1 part b. Students may miss key pieces of the explanation. For example, students may say that the temperature decreases by 5 degrees each year. It's critical that the language about the 1-inch increase is included in the meaning.

2. Use inequalities to express an appropriate domain and range using the linear equation that models the data.

Answers may vary but should be reasonable given the data and knowledge of reasonable temperatures in Florida.

Domain	$0 \leq x \leq 6$
Range	$50 \leq y \leq 90$



Why do the inequality symbols include “or equal to” when expressing the domain and range?

3.7.2 Activity: Station 1: High Temperature and Rainfall (continued)

3. Use the model to predict the mean high temperature in January for a Florida city that has a mean of 4.0 inches of rainfall per year.

63.06°F

4. Suppose there is a city in Florida that averages 7.8 inches of rainfall per year.

- a. Use the equation to predict the mean high temperature in January for that city.

44.09°F

- b. Explain whether or not this January high temperature seems reasonable.

Answers will vary. It is reasonable. Since January is one of the colder months in Florida, it is reasonable to have an average temperature in the 40's. This is not a reasonable extrapolation. None of the rainfall data exceeds an average of 4.8 inches per year, so the correlation between rainfall and high temperature may no longer apply for a city with rainfall that far outside of the domain.

- c. Discuss with your partner if you think the model can be extrapolated to a city in Florida averaging 7.8 inches of rainfall per year. Summarize your discussion.

Answers will vary. Yes, because Florida shares similar weather characteristics, it would be reasonable. No it would not be reasonable because it's not clear what the effect on the temperatures may be if the average rainfall in a city per year increased that significantly.



For question 4 part b, students may say it's not reasonable, particularly in south Florida. Allow students to give a reasonable answer if it is justified.

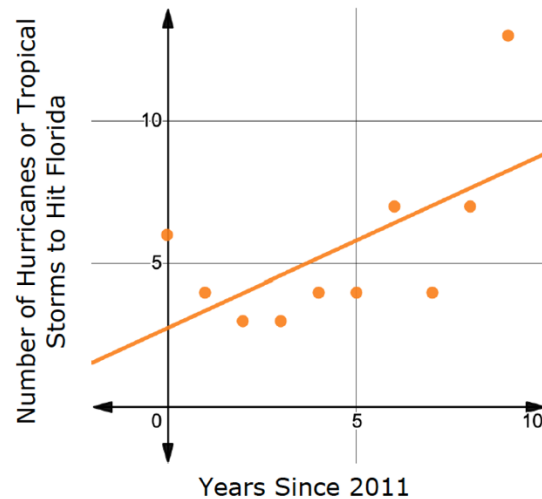
3.7.3 Activity: Station 2: Hurricanes in Florida Since 2011

Component Overview: Students will analyze the data regarding hurricanes and tropical storms in Florida to answer two, multipart questions.



3.7.3 Activity: Station 2: Hurricanes in Florida Since 2011

Michaela wondered if there was a trend between the number of hurricanes and tropical storms that hit Florida and time. She used data for the number of hurricanes and tropical storms to hit Florida from 2011-2020 to create the scatter plot shown.



1. The equation $y = 0.612x + 2.75$ models this relationship.
 - a. Interpret the meaning of the slope and y-intercept in context.

Value	Meaning
slope, m	<i>Each year since 2011, the number of hurricanes or tropical storms to hit Florida has increased by 0.612 storms per year, on average</i>
y-intercept, b	<i>The model predicts the number of hurricanes or tropical storms to hit Florida in 2011 was 2.75 storms</i>



Students can use the screenshots in the workbook or use the link in the digital area.



Listen for student responses to question 1 part b. Students may miss key pieces of the explanation. For example, students may not include the “since 2011” piece, which is critical to the interpretation.

3.7.3 Activity: Station 2: Hurricanes in Florida Since 2011 (continued)

- b. Explain why it does not make sense for the slope or y -intercept to be decimal values in this context.

There can't be a partial hurricane or tropical storm.

2. Use the model for the relationship, $y = 0.612x + 2.75$, to complete the following.

- a. Predict the number of hurricanes and tropical storms expected to hit Florida in 2023.

The model suggests there will be 10 hurricanes or tropical storms to hit Florida in 2023.

- b. Explain whether or not it seems reasonable for the number of hurricanes and tropical storms to hit Florida to continue increasing at this rate each year.

It is not reasonable to expect this. The number of storms won't continue to increase forever. This model simply represents the relationship between these two variables over a 10 year period and cannot necessarily be extrapolated outside of that period.

- c. Discuss with your partner how the model could be used to estimate the number of hurricanes and tropical storms that hit Florida in 2008. Summarize your discussion.

The value of -3 could be substituted for x in the model to estimate the number of hurricanes and tropical storms that hit Florida in 2008.



What would you do with data in this scenario if it came back with a decimal? Explain when you would round up or down.



If students finish this activity early, have them estimate the number of hurricanes in 2008 using the model, and then reveal that there were actually eight hurricanes or tropical storms to hit Florida that year. Then, have time for discussion on why the model may not be best to use for predictions and estimations outside of the domain of the data set.

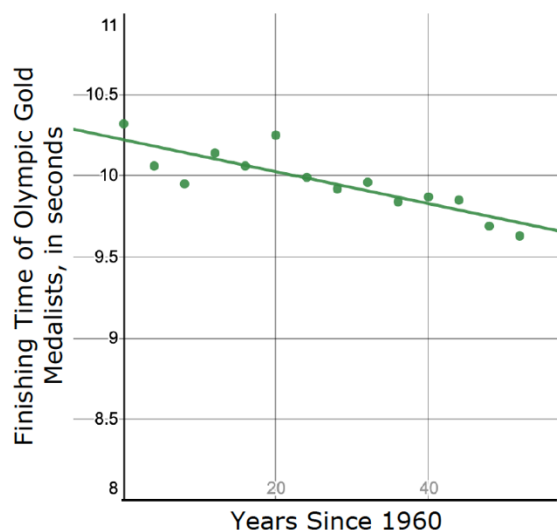
3.7.4 Activity: Station 3: Men's 100-Meter Dash at the Olympics

Component Overview: Students will analyze the data regarding the men's 100-meter dash at the Olympics since 1960 to answer two, multipart questions.



3.7.4 Activity: Station 3: Men's 100-Meter Dash at the Olympics

The scatter plot shows the finishing times for the Olympic gold medalist in the men's 100-meter (m.) dash for the Olympic games since 1960.



1. Dana says that the strength and direction of the linear association between the two variables is moderately negative. Explain whether or not you agree with Dana.

Answers will vary. I do not agree with Dana, because the values are closely aligned to the line of best fit, it would be a strong negative linear relationship.



Students can use the screenshots in the workbook or go to the actual link in the digital area.



MA.K12.MTR.4.1: Discussion and Reflection
Question 1 provides the opportunity for students to engage with this MTR, as students will analyze the mathematical thinking of others and discuss their thinking with peers.

3.7.4 Activity: Station 3: Men's 100-Meter Dash at the Olympics (continued)

2. The equation of the line of best fit shown on the scatter plot is $f(x) = 10.2217 - 0.0098x$, where f is the finishing time of the Olympic gold medalist in the men's 100-m dash and x is the number of years since 1960.
 - a. Interpret the y -intercept of the linear equation modeling the data.
The model predicts the finishing time of the Olympic gold medalist in the men's 100-m dash in 1960 was 10.2217 seconds.
 - b. Discuss with your partner the difference between the finishing time for the men's gold medal winner for the 100-m dash in the 1960 Olympics and the y -intercept of the linear equation.
The actual finishing time was slightly slower than the predicted time using the model.
 - c. Interpret the slope of the linear equation that models the data.
The finishing time since 1960 has decreased each year by 0.0098 seconds.
 - d. The summer Olympic games, where the 100-m dash is run, occur every 4 years. By how many seconds does the model predict the first place finishing time will decrease every 4 years?
0.0392
 - e. Discuss with your partner whether or not it is reasonable to use this model to predict the first place finishing time in the men's 100-m dash for the 2050 summer Olympics. Summarize your discussion.
Answers will vary. It isn't reasonable because eventually it won't be at the speed of a run and that speed may not qualify for the Olympics.



Not all students will recognize the correct answer to question 2 part b. Listen to their ideas to see if they are making connections between the model being used for estimates being different from the actual data values, as this will be the focus of the next lesson.

3.7.5 Wrap-Up: Growth Mindset

Component Overview: Students will identify their favorite mistake from this lesson.



3.7.5 Wrap-Up: Growth Mindset

1. Mistakes are an opportunity to learn. Complete the statement to highlight a mistake you learned from.

Means

I

Start

To

Acquire

Knowledge

Experience

Skills

My favorite mistake from today was ...

Answers will vary.



The information provided from the student here, coupled with what was observed during the lesson activities, can provide information for differentiating instruction in the next lesson.